

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	7 March 2026
Team ID	
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Image Upload	Upload blood cell image (JPG/PNG) Preview uploaded image before prediction Validate image format and size
FR-4	Malaria Detection Prediction	Analyze uploaded blood cell image using AI model Predict result as <i>Parasitized</i> or <i>Uninfected</i> Display prediction result to the user

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have a simple and user-friendly interface so that doctors and lab technicians can easily upload images and view results.
NFR-2	Security	User data and uploaded images must be protected using secure authentication and data encryption to ensure privacy

NFR-3	Reliability	The system should provide accurate and consistent predictions for malaria detection using a trained AI model.
NFR-4	Performance	The system should process uploaded images and generate prediction results within a few seconds.
NFR-5	Availability	The application should be available online and accessible anytime for healthcare professionals.
NFR-6	Scalability	The system should handle multiple users and image uploads at the same time without reducing performance.